VEDA Technology Internship — Task 26 Report

Signup Form with Validation Documentation & Comprehensive Verification (Web Development Track)

1. Project Overview

This report documents the design, architecture, and verification of **Task 26 (Signup Form with Validation)** for the VEDA Technology Web Development Internship. The project delivers an accessible, robust, and responsive user registration frontend engineered strictly with **HTML5**, **CSS3**, and **Vanilla JavaScript (ES6+)**. Styled with an elegant **Light Pastel Green and White** theme (#F4F8F5 background, #FFFFFF card, #2D6A4F forest green accent, #EAF5EE badge tints, #D5E5DA borders), the application features real-time and submit-time validation for Name, Email, and Password, RFC-compliant email regex testing, live 3-tier password strength evaluation (Weak, Medium, Strong), an interactive password complexity checklist, Show/Hide password toggle, instant error clearing upon input correction, and an accessible verified success screen with zero external libraries or backend dependencies.

2. Technical Architecture & Component Stack

Component	Implementation & Technical Details
HTML5 Semantic Architecture	Structured with <main>, <form novalidate>, accessible <label for="">, aria-required, aria-describedby, and aria-live="polite".
Light Pastel Green Design System	CSS custom properties with white surfaces, soft mint tints, high-contrast accessible text, subtle shadows, and 3px focus rings.
Name Format & Bounds Guard	Regex /^[A-Za-z\s'-]{2,50}$/ guaranteeing names are non-empty, ≥ 2 characters, and restricted to alphabetic characters, spaces, hyphens, and apostrophes.
RFC Email Validation Regex	Client-side RFC-compliant regular expression validating username, domain, and TLD syntax with instant error reporting.
Password Length & Complexity	Multi-factor criteria enforcement requiring ≥ 8 characters, uppercase, lowercase, numbers, and special symbols.
Live Password Strength Meter	Dynamic 3-segment visual meter and color-coded status badges (Weak: orange, Medium: amber, Strong: pastel green) calculated on keystroke.
Live Complexity Checklist	Interactive 5-item bullet list dynamically switching to green checkmarks as individual password conditions are satisfied.
Show / Hide Password Toggle	Accessible toggle button switching input type between password and text with synchronized eye/eye-slash SVG icons.
Zero Dependencies & GitHub Ready	100% lightweight vanilla code with zero frameworks, optimized for instant load times and GitHub Pages deployment.

3. Functional Specifications Breakdown

Requirement	Logic & Implementation Description	Status
1. Name, Email, Password Fields	Semantic inputs with explicit labels, placeholders, and autocomplete attributes.	**PASSED**
2. Full Name Validation	Guarantees non-empty input, minimum length of 2 characters, and valid name format.	**PASSED**
3. Email Format Validation	Evaluates address syntax against RFC standard regex pattern; catches missing @ or domain.	**PASSED**
4. Password Complexity Rules	Validates ≥ 8 chars, uppercase, lowercase, digit, and special symbol with helpful feedback.	**PASSED**
5. Inline Error Messages	Renders clear, accessible error messages below relevant fields; no browser alerts.	**PASSED**
6. Password Strength Indicator	Live 3-segment progress indicator dynamically classifying strength as Weak, Medium, or Strong.	**PASSED**
7. Form Submission Prevention	Calls preventDefault() on submit; halts submission and shifts focus to first invalid input.	**PASSED**
8. Verified Success Screen	Transitions smoothly to a personalized success screen summarizing verified user data.	**PASSED**

9. Show / Hide Password Button	Toggles password visibility with accessible button and dual eye SVG icons.	**PASSED**
10. Live Error Clearing	Immediately clears error styling and messages on input event as user corrects errors.	**PASSED**

4. Verification & Testing Matrix

Test Scenario	Input Action	Expected State	Actual State	Status
Empty Form Submit	Click "Create Account" blank	3 inline errors show; focus on Name input	All 3 errors render; focus moved to Name	**PASSED**
Invalid Name String	Enter "J" or "Jane123"	Inline error indicates length or letter rules	Clear error message displayed below name	**PASSED**
Valid Name String	Enter "Jane Doe"	Error clears; green valid indicator appears	Error removed; valid border and checkmark set	**PASSED**
Invalid Email Format	Enter "jane.doe@" or "test"	Shows "Please enter a valid email address..."	Accurate syntax error displayed inline	**PASSED**
Valid Email Format	Enter "jane@example.com"	Error clears; input marked valid	Error clears instantly; green check displayed	**PASSED**
Short / Weak Password	Enter "pass"	Strength badge "Weak", 1 orange segment	Meter shows Weak; unfulfilled items active	**PASSED**
Medium Password	Enter "Password12"	Strength badge "Medium", 2 amber segments	Meter shows Medium; special symbol missing	**PASSED**
Strong Password	Enter "Secure#2026Pass"	Strength badge "Strong", 3 green segments	All 5 checklist items turn green; Strong badge	**PASSED**
Show/Hide Toggle	Click eye toggle button	Toggles type "password" <-> "text" with icon update	Text obscured/revealed cleanly; focus retained	**PASSED**
Successful Signup	Submit all valid credentials	Form hides; success card displays user name	Personalized success screen shown; reset ready	**PASSED**

5. Deployment & Repository Structure

- **Directory:** Task26-Signup-Form-Validation/
- **Files:** index.html, style.css, script.js, README.md
- **GitHub Repository:** https://github.com/dharmarayank946/Signup-Form-Validation.git
- **Default Branch:** main (Commit: 9d08f04)
- **Developer:** Nagalambika D Koralli • VEDA Technology Internship